PROFINITE SETS IN PRINCIPAL AFFINES UNDER LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. No omissions: the reduction is worked and the finite engine is a proved same-chapter lemma.

Allowed standard prerequisites (editorial).

- Finite-set principal affine lemma. Limits, finite set in principal affine lemma (same chapter)

Source reference. Original reference 01PS: sources/stacks-chow/tags.txt, source lines 2241–2241. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01PY: sources/stacks-chow/tags.txt, source lines 2247–2247. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09MT: sources/stacks-chow/tags.txt, source lines 12191–12191. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B8W: sources/stacks-chow/tags.txt, source lines 14154–14154. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZC: sources/stacks-chow/tags.txt, source lines 2587–2587. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09NV: sources/stacks-chow/tags.txt, source lines 12228–12228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{L}$ be an ample invertible sheaf on X. Assume we have morphisms of schemes*

$$\operatorname{Spec}(k) \leftarrow \operatorname{Spec}(A) \to W \subset X$$

where k is a field, A is an integral k-algebra, W is open in X. Then there exists an $n > 0$ and a section $s \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that X_s is affine, $X_s \subset W$, and $\operatorname{Spec}(A) \to W$ factors through X_s

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Proof. Since $\mathrm{Spec}(A)$ is quasi-compact, we may replace W by a quasi-compact open still containing the image of $\mathrm{Spec}(A) \to X$. Recall that X is quasi-separated and quasi-compact by dint of having an ample invertible sheaf, see Properties, Definition 01PS and Lemma 01PY. By Proposition 01ZA we can write $X = \lim X_i$ as a limit of a directed system of schemes of finite type over $\mathbf{Z}$ with affine transition morphisms. For some i the ample invertible sheaf $\mathcal{L}$ on X descends to an ample invertible sheaf $\mathcal{L}_i$ on X_i and the open W is the inverse image of a quasi-compact open $W_i \subset X_i$, see Lemmas 09MT, 0B8W, and 01Z4. We may replace $X, W, \mathcal{L}$ by $X_i, W_i, \mathcal{L}_i$ and assume X is of finite presentation over $\mathbf{Z}$. Write $A = \mathrm{colim}\, A_j$ as the colimit of its finite k-subalgebras. Then for some j the morphism $\mathrm{Spec}(A) \to X$ factors through a morphism $\mathrm{Spec}(A_j) \to X$, see Proposition 01ZC. Since $\mathrm{Spec}(A_j)$ is finite this reduces the lemma to Properties, Lemma 09NV. $\square$